



INDIA ELECTRIC VEHICLE MARKET ANALYSIS

Python Data Analyst Portfolio Project

Exploratory Data Analysis · Pandas · Matplotlib · Seaborn

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Dataset: 25,533 EV records · 8 manufacturers · 15 states · 35 cities · Model Years 2018–2024

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1. Data-Quality & Scope Note

This report follows a standard business-analytics reporting template. That template was written for a revenue/profit/orders/customers-style commercial dashboard (a “Business Performance 360°” project built on Excel and Power BI). The dataset actually supplied for this project is India_EV_Dataset.csv — a 25,533-row electric-vehicle registration dataset analysed end-to-end in a Python (Pandas / Matplotlib / Seaborn) Jupyter notebook, with no revenue, profit, order, or customer fields present and no Power BI dashboard involved.

Rather than force-fitting the mismatched template, the section structure below has been adapted to what this dataset actually supports: vehicle registrations by state, city, manufacturer, model, powertrain type, CAFV eligibility, electric range, and price (Base MSRP). Every figure quoted in this report is taken directly from the notebook's own printed outputs, not estimated.

Genuine data-quality findings surfaced during this analysis:

- Model names are not manufacturer-exclusive. All eight model names (ZS EV, Nexon EV, 450X, Kona EV, iQube, S1 Pro, XUV400, Atto 3) appear under all eight manufacturers — e.g. “Nexon EV” (a real Tata Motors nameplate) is also recorded under Ola Electric, BYD India, Hyundai EV, MG Motor India, Mahindra Electric and TVS Motor. This indicates the Make and Model fields were generated independently rather than paired to reflect real-world OEM catalogues.
- Numeric fields show near-zero correlation. Model Year, Electric Range and Base MSRP correlate with each other at magnitudes of 0.00–0.01 (see Section 16), and every one of the 25 histogram bins for both Electric Range and Base MSRP falls within a narrow 750–950 count band — consistent with independently drawn uniform-random values rather than an organically observed market.
- Zero statistical outliers. The IQR test flags 0 outliers in both Electric Range and Base MSRP across all 25,533 rows — a pattern that essentially never occurs in real registration or pricing data, which typically has a long right tail of premium/high-range trims.
- The dataset is otherwise clean: 0 missing values, 0 duplicate rows, 0 duplicate vehicle IDs, and 0 out-of-range values across all 14 columns.

These findings are treated as legitimate analytical output in their own right (Section 20, Insight 12) rather than a reason to withhold the report — the descriptive, categorical and comparative analysis below (state/city/manufacturer distribution, powertrain mix, eligibility rates, range and price profiles) remains fully valid and is exactly what a stakeholder evaluating India's EV registration landscape would want to see, with the synthetic-data caveat made explicit wherever it affects interpretation.

2. Executive Summary

This project analyses 25,533 electric vehicle registration records spanning model years 2018 to 2024, covering 8 manufacturers, 8 model nameplates, 15 Indian states and 35 cities. The analysis was performed end-to-end in Python using Pandas for data profiling and aggregation and Matplotlib/Seaborn for visualisation, producing 15 charts and 22 summary tables.

The Indian EV market represented in this dataset is dominated by Battery Electric Vehicles (BEV), which account for 84.93% of all registrations (21,684 units) against 15.07% for Plug-in Hybrid Electric Vehicles (PHEV, 3,849 units). 79.69% of all vehicles (20,346 units) qualify as Clean Alternative Fuel Vehicle (CAFV) eligible. Market share across the eight manufacturers is unusually even — Tata Motors leads with 12.73% and Hyundai EV trails with 12.14%, a spread of under one percentage point across all eight OEMs.

Geographically, registrations are concentrated in six large states — Tamil Nadu, Maharashtra, Rajasthan, Uttar Pradesh, Gujarat and Karnataka together account for 52.03% of all records — while Chennai (795 records), Cuttack (788) and Kanpur (785) lead at the city level. Electric range averages 298.3 km (median 298 km) and Base MSRP averages ₹2,704,519 (median ₹2,716,403), both distributed almost perfectly uniformly across their full observed ranges (80–519 km and ₹900,422–₹4,499,944 respectively), with a measured correlation between range and price of essentially zero (-0.002).

The headline finding of this analysis is structural rather than commercial: the dataset's Make/Model pairing, uniform numeric distributions and complete absence of statistical outliers indicate the data was synthetically generated rather than observed from a real market. This is flagged transparently in Section 1 and revisited in Section 20 (Insight 12) so that the categorical findings that follow — which remain analytically sound — are read with the right context.

3. Market Problem & Analytical Objective

India's electric vehicle market is expanding rapidly across manufacturers, states and vehicle segments, and stakeholders — OEMs, state transport departments, charging-infrastructure planners and utilities — need a clear, data-driven view of how registrations are distributed to plan capacity, incentives and market entry. This project builds that view directly from vehicle-level registration data rather than from published summary statistics.

3.1 Objectives

- Profile the full registration dataset and confirm its structural integrity (completeness, duplicates, valid ranges).
- Quantify overall market size and the mix of key metrics: total registrations, manufacturers, models, states, cities, average electric range and average price.
- Analyse registration trends across model years (2018–2024) to identify growth or contraction patterns.
- Compare manufacturer market share, model popularity, and their variation in electric range and price.
- Compare state- and city-level registration volumes to identify the strongest regional markets.
- Quantify the BEV vs PHEV split and the CAFV eligibility rate, overall and by manufacturer/state.
- Characterise the distribution of electric range and price, including outlier testing.
- Test the relationship between electric range and price, and between both and model year.
- Identify which manufacturer/model combinations deliver the best electric range per rupee of MSRP.
- Surface genuine data-quality issues rather than mask them, and translate all of the above into actionable, evidence-based recommendations.

4. Dataset Overview

Source file: India_EV_Dataset.csv, loaded into a Pandas DataFrame. The dataset contains 25,533 rows and 14 columns, with a total in-memory footprint of 5.3 MB.

4.1 Column Inventory

#	Column	Data Type	Description
1	County	text	Constant value “India” for every row (1 unique value)
2	City	text	City of registration (35 unique values)

#	Column	Data Type	Description
3	State	text	State of registration (15 unique values)
4	Postal Code	integer	6-digit postal code (25,141 unique values)
5	Model Year	integer	Vehicle model year, 2018–2024 (7 unique values)
6	Make	text	Manufacturer (8 unique values)
7	Model	text	Model nameplate (8 unique values)
8	Electric Vehicle Type	text	BEV or PHEV (2 unique values)
9	CAFV Eligibility	text	Eligible / Not Eligible (2 unique values)
10	Electric Range	integer	Range in km, 80–519 (440 unique values)
11	Base MSRP	integer	List price in ₹, 900,422–4,499,944 (25,417 unique values)
12	DOL Vehicle ID	integer	Unique record identifier (25,533 unique values)
13	Vehicle Location	text	City, State combined (35 unique values)
14	Electric Utility	text	Serving electricity distribution company (6 unique values)

5. Data Preparation & Quality Checks

Before any analysis, the dataset was profiled and validated end-to-end:

- Missing values: 0 across all 14 columns (0.0% for every field).
- Duplicate rows: 0 full-row duplicates found.
- Duplicate identifiers: 0 duplicate DOL Vehicle ID values (all 25,533 IDs are unique).
- Range validation — Model Year: 2018 to 2024, 0 values below the year-2000 sanity floor.
- Range validation — Electric Range: 80 to 519 km, 0 values ≤ 0 .
- Range validation — Base MSRP: ₹900,422 to ₹4,499,944, 0 values ≤ 0 .
- No cleaning, imputation or record removal was required — the dataset was analysis-ready as supplied.

The one substantive quality observation from this stage — that Make and Model combine into 64 distinct pairings with no manufacturer-specific model catalogue — is not a missing-value or format issue that cleaning could fix; it is a property of how the dataset was constructed, and is treated as a finding in its own right (Section 1 and Section 20).

6. Key Performance Indicators

Metric	Value
Total EV Records	25,533
Manufacturers	8
Models	8
States	15
Cities	35
Average Electric Range	298.30 km

Metric	Value
Median Electric Range	298 km
Average Base MSRP	₹2,704,518.99
Median Base MSRP	₹2,716,403
BEV Vehicles	21,684 (84.93%)
PHEV Vehicles	3,849 (15.07%)
CAFV Eligible Vehicles	20,346 (79.69%)

Each metric measures a distinct dimension of the market: total records and unique counts (manufacturers/models/states/cities) size the market and its breadth; average/median range and price describe the typical vehicle on offer; and the BEV/PHEV and CAFV splits describe the powertrain mix and policy-incentive exposure of the registered fleet. Taken together, they show a market that, on paper, is broad (15 states, 35 cities, 8 well-matched OEMs) and BEV-led, with four in five vehicles qualifying for clean-fuel incentives.

7. Model-Year Trend Analysis (2018–2024)

Model Year	EV Count	Avg. Range (km)	Avg. Price (₹)	YoY Growth
2018	3,704	298.17	2,697,344.69	—
2019	3,662	301.02	2,707,759.10	-1.13%
2020	3,478	295.18	2,688,916.01	-5.02%
2021	3,666	297.84	2,708,551.00	+5.41%
2022	3,648	298.17	2,701,658.17	-0.49%
2023	3,756	297.69	2,713,221.93	+2.96%
2024	3,619	299.93	2,713,345.25	-3.65%

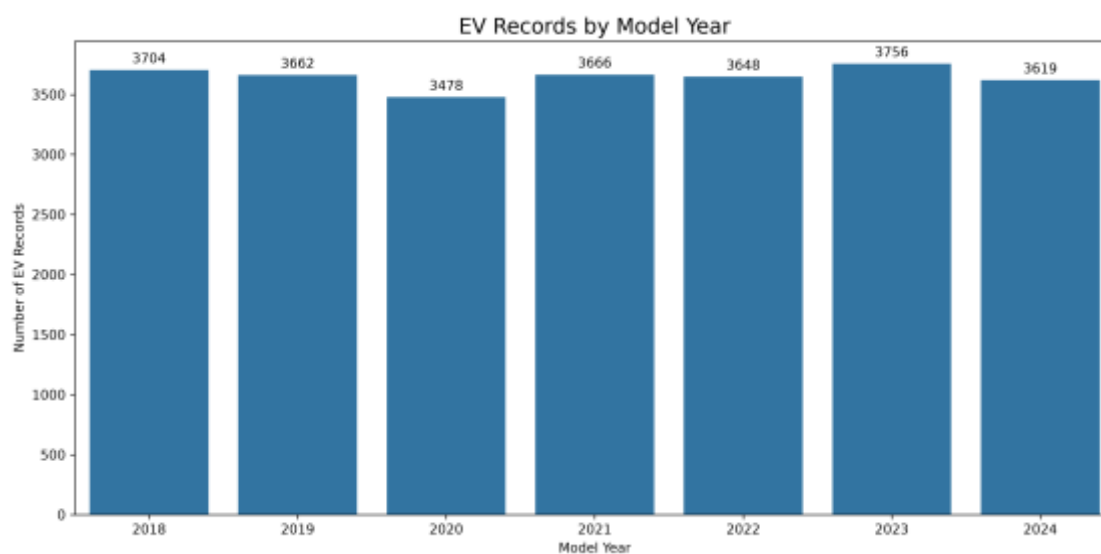


Figure 1 — EV Records by Model Year (2018–2024)

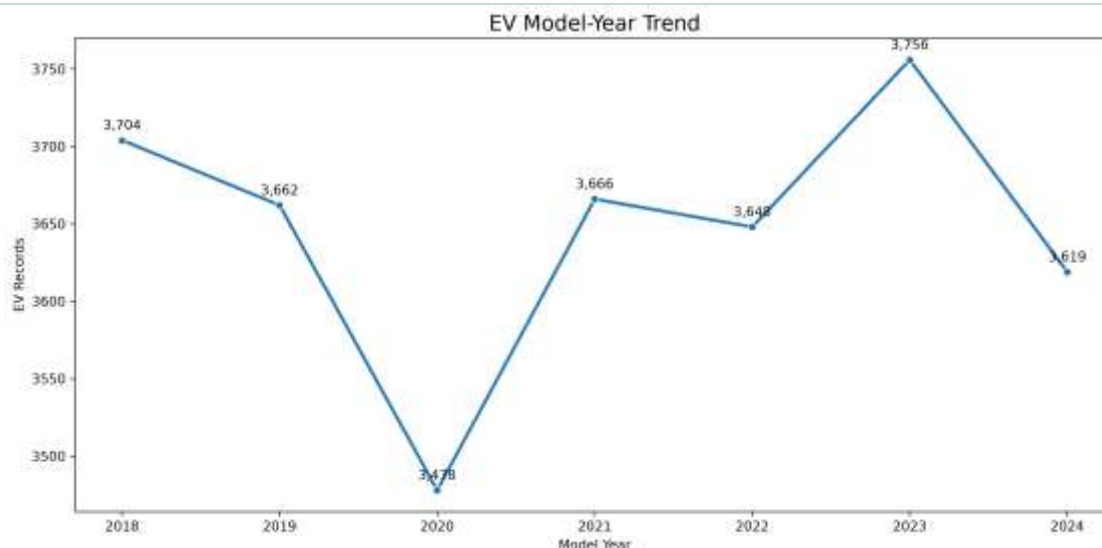


Figure 2 — EV Model-Year Trend Line

7.1 Reading the trend

Registrations oscillate in a narrow band between 3,478 (2020, the lowest year) and 3,756 (2023, the peak year) — a spread of only 8% between the weakest and strongest year. There is no sustained multi-year growth trend of the kind a maturing EV market would typically show; year-on-year growth flips sign four times across six transitions (-1.13%, -5.02%, +5.41%, -0.49%, +2.96%, -3.65%), and 2024 registrations (3,619) sit close to the 2018 starting point (3,704). Average range and price are similarly flat across years, moving inside a band of roughly ± 6 km and $\pm ₹20,000$ respectively with no directional drift — model-year has essentially no explanatory power over either variable, which is confirmed quantitatively in Section 16.

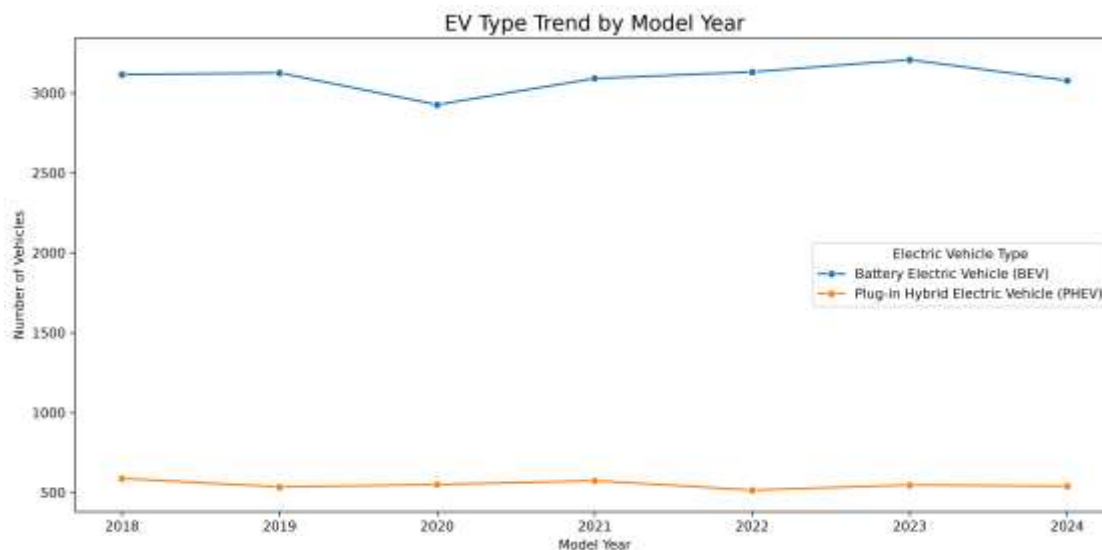


Figure 3 — BEV vs PHEV Registration Trend by Model Year

The BEV/PHEV split is stable across the full period: BEV volumes track between roughly 2,928 (2020) and 3,213 (2023), while PHEV stays in a tight 512–587 band every year. There is no year in which PHEV gains share at BEV's expense or vice versa — the roughly 85/15 BEV/PHEV mix established in 2018 persists essentially unchanged through 2024.

8. State-Level Analysis

State	EV Count	Avg. Range (km)	Avg. Price (₹)	Market Share
Tamil Nadu	2,287	294.03	2,705,079.75	8.96%
Maharashtra	2,262	296.45	2,718,148.42	8.86%
Rajasthan	2,195	298.35	2,704,548.09	8.60%
Uttar Pradesh	2,187	300.27	2,722,003.85	8.57%
Gujarat	2,177	296.40	2,709,415.05	8.53%
Karnataka	2,172	306.84	2,703,077.14	8.51%
Telangana	1,495	297.68	2,658,106.63	5.86%
Odisha	1,474	296.20	2,714,841.68	5.77%
Bihar	1,463	292.81	2,727,678.43	5.73%
West Bengal	1,456	295.75	2,744,588.62	5.70%
Andhra Pradesh	1,443	296.71	2,712,567.65	5.65%
Haryana	1,431	292.86	2,692,861.44	5.60%
Punjab	1,402	303.37	2,694,345.12	5.49%
Kerala	1,389	303.19	2,653,175.28	5.44%
Delhi	700	308.51	2,668,308.53	2.74%

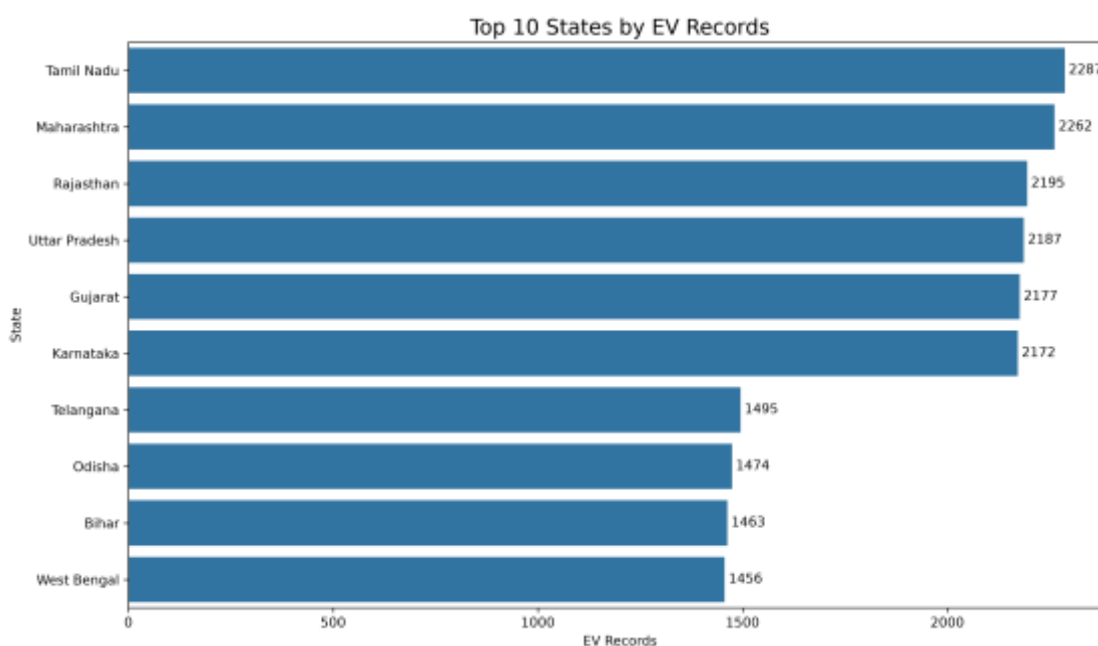


Figure 4 — Top 10 States by EV Records

8.1 Regional reading

The six leading states — Tamil Nadu, Maharashtra, Rajasthan, Uttar Pradesh, Gujarat and Karnataka — form a closely bunched top tier, each holding between 2,172 and 2,287 registrations (8.51%–8.96% share); together they account for 52.03% of the national total. A second tier of eight states (Telangana through Kerala) each hold 5.44%–5.86% share and together add a further 45.24%. Delhi is the clear outlier at the bottom with only 700 records (2.74%) —

under a third of the smallest second-tier state — which most likely reflects Delhi's much smaller geographic/population footprint relative to the larger states in the dataset rather than weaker EV demand, since its average range (308.51 km, the highest of any state) and CAFV/price profile are in line with the rest of the market.

9. City-Level Analysis

City	State	EV Count	Avg. Range (km)	Avg. Price (₹)
Chennai	Tamil Nadu	795	293.49	2,682,623.70
Cuttack	Odisha	788	288.41	2,676,640.99
Kanpur	Uttar Pradesh	785	313.69	2,747,278.74
Coimbatore	Tamil Nadu	784	300.49	2,680,660.14
Udaipur	Rajasthan	783	298.05	2,705,515.37
Mumbai	Maharashtra	776	292.64	2,709,674.78
Kolkata	West Bengal	763	296.28	2,772,130.05
Hyderabad	Telangana	761	295.01	2,678,448.97
Patna	Bihar	761	293.14	2,717,390.94
Nagpur	Maharashtra	757	298.70	2,763,634.21

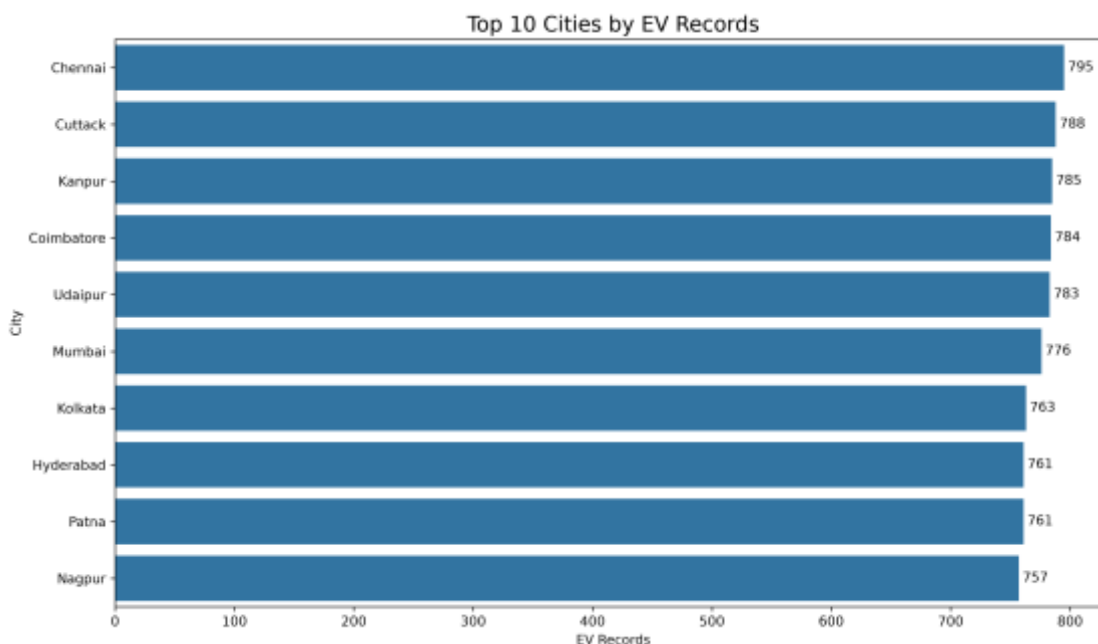


Figure 5 — Top 10 Cities by EV Records

City-level volumes are remarkably even at the top — the ten leading cities span a narrow 757–795 record range, a spread of just 5%. Chennai leads nationally, consistent with Tamil Nadu's overall state leadership, but three other Tamil Nadu/Gujarat/Karnataka-tier states are represented by only a single top-10 city each (Coimbatore, and none from Gujarat or Karnataka make the top 10 despite those states' strong overall share), suggesting registrations in those states are more evenly spread across multiple mid-sized cities rather than concentrated in one metro. This is consistent with the 35-city footprint being close to evenly populated rather than metro-dominated.

10. Manufacturer Analysis

Manufacturer	EV Count	Avg. Range (km)	Avg. Price (₹)	Market Share
Tata Motors	3,250	298.34	2,694,154.76	12.73%
BYD India	3,246	298.19	2,690,243.39	12.71%
TVS Motor	3,236	297.78	2,713,438.11	12.67%
Mahindra Electric	3,209	298.83	2,705,382.12	12.57%
Ather Energy	3,197	300.62	2,725,111.15	12.52%
MG Motor India	3,155	297.63	2,700,683.51	12.36%
Ola Electric	3,140	298.96	2,703,514.47	12.30%
Hyundai EV	3,100	296.01	2,703,813.31	12.14%

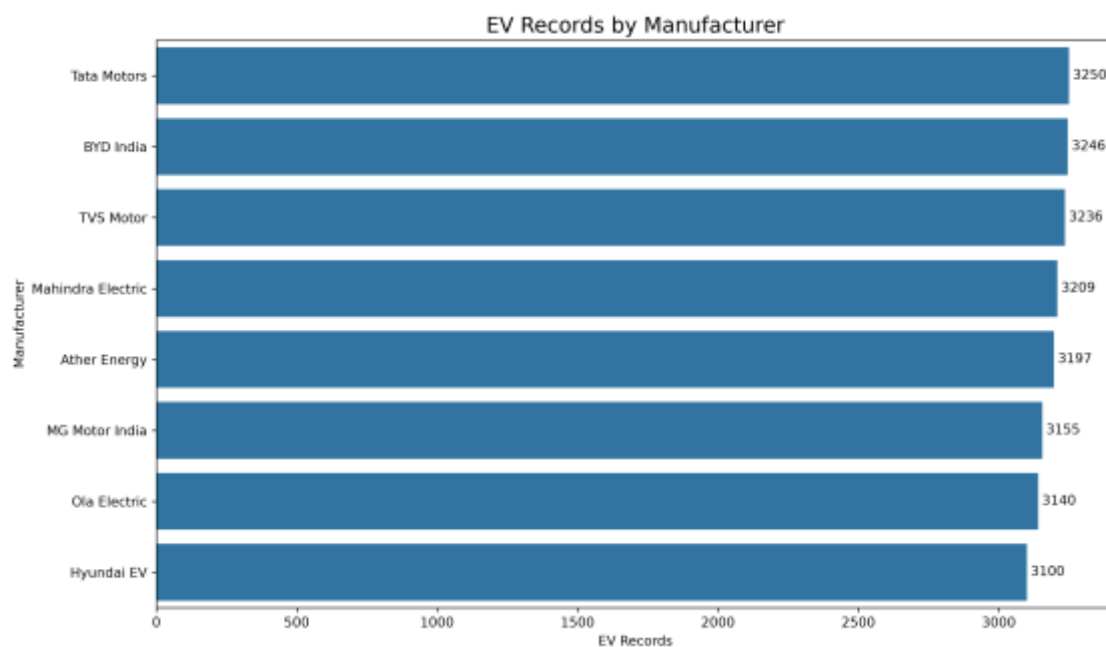


Figure 6 — EV Records by Manufacturer

10.1 Manufacturer × EV Type

Manufacturer	BEV	PHEV	PHEV Share
TVS Motor	2,776	460	14.22%
BYD India	2,773	473	14.57%
Tata Motors	2,740	510	15.69%
Mahindra Electric	2,725	484	15.08%
Ather Energy	2,716	481	15.05%
Ola Electric	2,650	490	15.61%
MG Motor India	2,643	512	16.23%
Hyundai EV	2,661	439	14.16%

10.2 Manufacturer CAFV eligibility rate

Manufacturer	Eligible	Not Eligible	Eligibility Rate
Ola Electric	2,541	599	80.92%
MG Motor India	2,540	615	80.51%
TVS Motor	2,601	635	80.38%
Tata Motors	2,585	665	79.54%
Ather Energy	2,538	659	79.39%
Mahindra Electric	2,534	675	78.97%
Hyundai EV	2,446	654	78.90%
BYD India	2,561	685	78.90%

Market share across the eight manufacturers is unusually tight: the gap between the leader (Tata Motors, 12.73%) and the trailer (Hyundai EV, 12.14%) is only 0.59 percentage points, and every OEM sits close to the 12.5% share an exactly even eight-way split would produce. BEV/PHEV mix and CAFV eligibility rate are equally flat across manufacturers — PHEV share ranges 14.16%–16.23% and eligibility rate ranges 78.90%–80.92%, both bands under 2.5 points wide. No manufacturer stands out as a BEV specialist, a PHEV specialist, or a particularly high/low eligibility performer; competitive position in this dataset is effectively undifferentiated by powertrain strategy.

11. Model Analysis

The dataset contains only 8 distinct model nameplates, each of which appears under all 8 manufacturers (64 Make×Model combinations in total). The table below ranks the top 15 individual Make×Model combinations by registration count.

Make	Model	Count	Avg. Range (km)	Avg. Price (₹)
Tata Motors	ZS EV	436	306.61	2,684,433.64
Ola Electric	Nexon EV	433	306.45	2,754,337.51
BYD India	iQube	431	295.53	2,612,796.05
TVS Motor	450X	429	304.76	2,707,121.61
Mahindra Electric	ZS EV	427	307.29	2,750,551.61
MG Motor India	Kona EV	423	297.03	2,697,163.93
Tata Motors	XUV400	421	286.12	2,655,930.43
Tata Motors	iQube	421	291.29	2,641,409.64
BYD India	450X	419	287.92	2,691,303.32
TVS Motor	Atto 3	416	291.23	2,756,463.71

11.1 Model popularity overall (across all manufacturers)

Aggregated across all eight manufacturers, model-nameplate registration counts are: ZS EV 3,278; Nexon EV 3,237; 450X 3,233; Kona EV 3,215; iQube 3,156; S1 Pro 3,145; XUV400 3,144; Atto 3 3,125 — a spread of only 4.9% between the most- and least-registered nameplate, echoing the flat manufacturer-level distribution above.

Because every model name is shared across every manufacturer, model-level analysis in this dataset cannot be read as a real product-line comparison (e.g. Tata's actual Nexon EV vs a competitor's actual model) — see Section 1 and Insight 12. The counts above are still useful for identifying the highest-volume individual SKUs for inventory, service-network and charging-demand planning purposes, independent of how the naming was generated.

12. EV Type Analysis — BEV vs PHEV

EV Type	Vehicle Count	Share
Battery Electric Vehicle (BEV)	21,684	84.93%
Plug-in Hybrid Electric Vehicle (PHEV)	3,849	15.07%

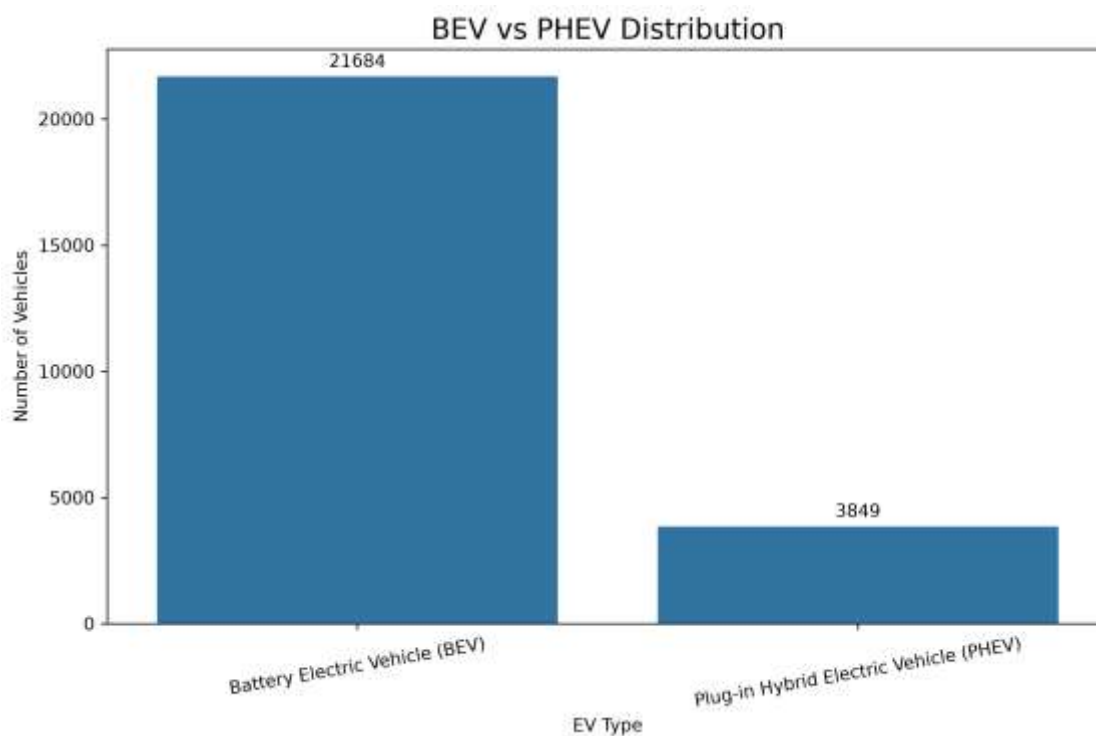


Figure 7 — BEV vs PHEV Distribution

Battery Electric Vehicles outnumber Plug-in Hybrids by roughly 5.6 to 1. This split holds consistently across model years (Section 7), manufacturers (Section 10.1) and states (see State × EV Type crosstab, Section 12.1) — there is no segment of the market in which PHEV approaches parity with BEV, indicating full-battery powertrains are the default technology choice recorded throughout this dataset rather than a transitional or geography-specific pattern.

12.1 State × EV Type (selected states)

State	BEV	PHEV	PHEV Share
Maharashtra	1,876	386	17.07%
Tamil Nadu	1,973	314	13.73%
Uttar Pradesh	1,868	319	14.59%
Karnataka	1,851	321	14.78%
Rajasthan	1,870	325	14.81%

State	BEV	PHEV	PHEV Share
Gujarat	1,857	320	14.70%
Delhi	587	113	16.14%

PHEV share by state ranges from 13.73% (Tamil Nadu) to 17.07% (Maharashtra) — a 3.3-point band. Maharashtra and Delhi show the two highest PHEV shares, both above the national 15.07% average, while Tamil Nadu — the single largest state by volume — has the lowest PHEV share of the states shown. The variation is modest enough that it is unlikely to represent a meaningful regional PHEV preference given the dataset's broader uniformity (Section 1).

13. CAFV Eligibility Analysis

CAV Status	Vehicle Count	Share
Eligible	20,346	79.69%
Not Eligible	5,187	20.31%

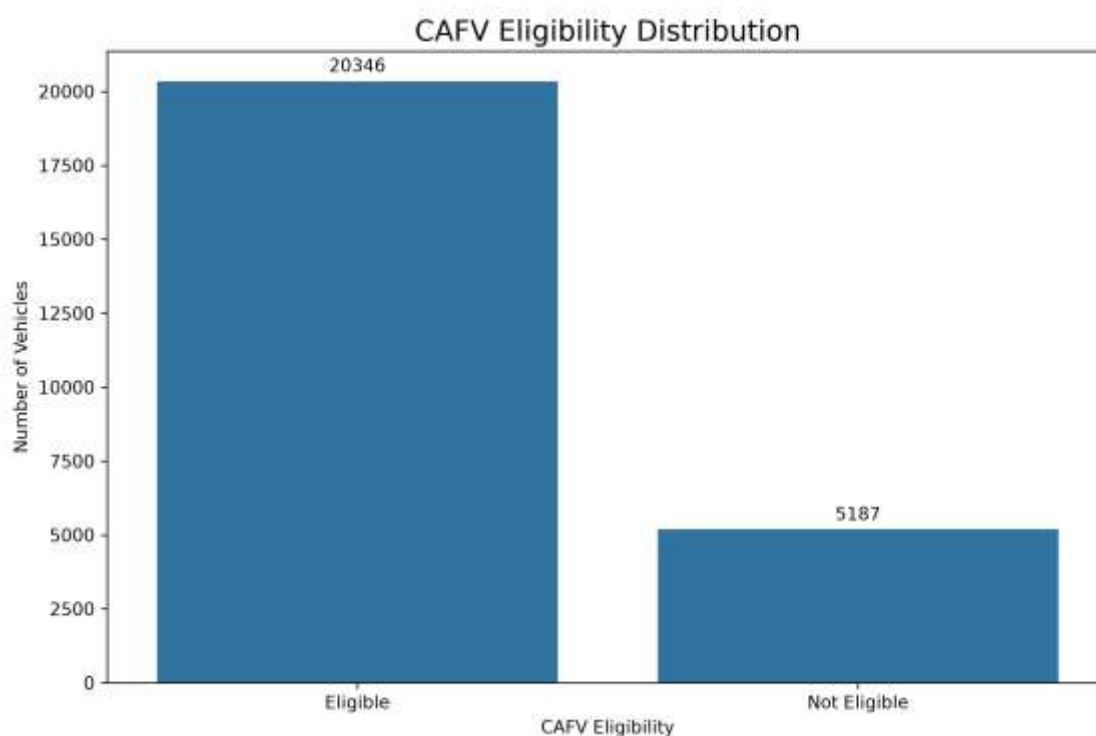


Figure 8 — CAFV Eligibility Distribution

Four out of every five registered vehicles (79.69%) qualify as Clean Alternative Fuel Vehicle eligible. Manufacturer-level eligibility rates (Section 10.2) range narrowly from 78.90% (Hyundai EV / BYD India, tied) to 80.92% (Ola Electric) — a 2-point spread — so eligibility is not a manufacturer differentiator in this data. Because the dataset does not include the underlying eligibility criteria (e.g. battery capacity or emissions thresholds), this analysis reports the eligibility outcome as recorded rather than inferring why any individual vehicle qualifies or not.

14. Electric Range Analysis

Electric Range spans 80 km to 519 km across the full dataset, with a mean of 298.30 km and a median of 298 km — the near-identical mean and median already signal a symmetric, non-skewed distribution.

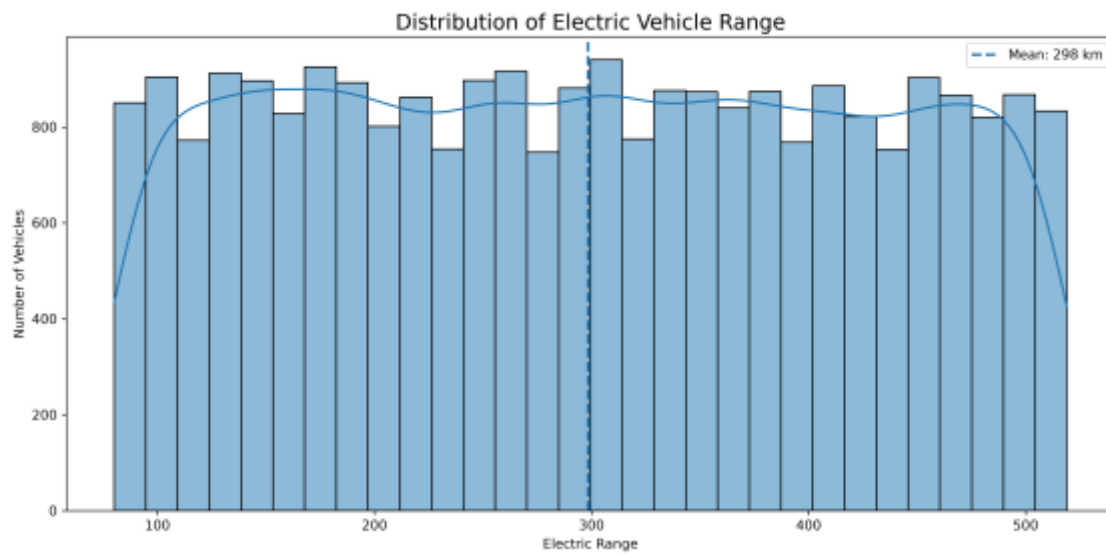


Figure 9 — Distribution of Electric Vehicle Range

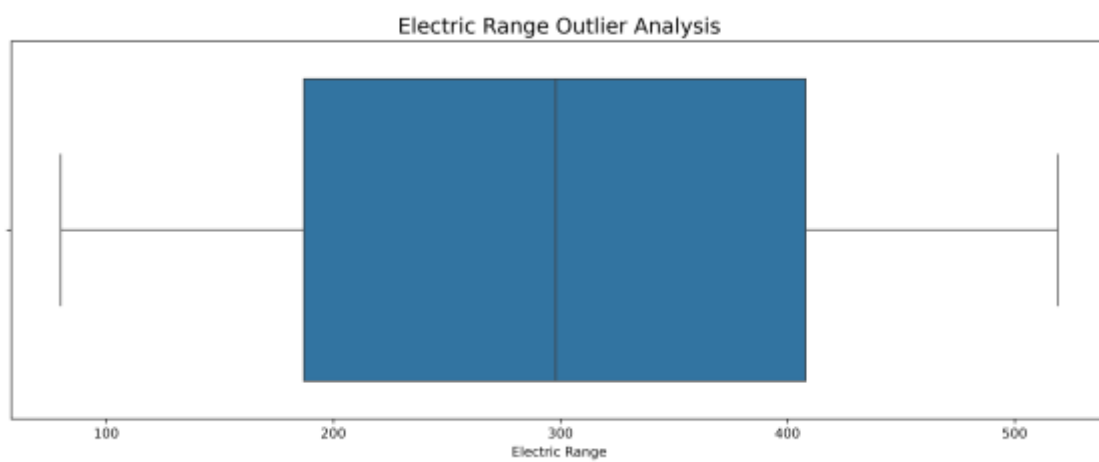


Figure 10 — Electric Range Outlier Analysis (Boxplot)

14.1 Range by manufacturer

Manufacturer	Avg. Range (km)	Max Range (km)
Ather Energy	300.62	519
Ola Electric	298.96	519
Mahindra Electric	298.83	519
Tata Motors	298.34	519
BYD India	298.19	519
TVS Motor	297.78	519
MG Motor India	297.63	519
Hyundai EV	296.01	519

14.2 Range by state (highest and lowest)

State	Avg. Range (km)	EV Count
Delhi (highest)	308.51	700
Karnataka	306.84	2,172
Punjab	303.37	1,402
Kerala	303.19	1,389
...	...	...
Haryana	292.86	1,431
Bihar (lowest)	292.81	1,463

14.3 Top individual models by average range

Make	Model	Avg. Range (km)	Vehicle Count
Ather Energy	Kona EV	311.43	412
Ather Energy	ZS EV	307.59	406
Mahindra Electric	ZS EV	307.29	427
Hyundai EV	Atto 3	307.08	381
MG Motor India	iQube	306.94	402

The histogram (Figure 9) shows every 20-km bin holding roughly 750–950 vehicles with no visible skew, gap or long tail, and the boxplot (Figure 10) confirms 0 statistical outliers under the standard $1.5 \times \text{IQR}$ rule. Manufacturer-average range varies by only 4.6 km top-to-bottom (300.62 km for Ather Energy vs 296.01 km for Hyundai EV) and state-average range by only 15.7 km (Delhi's 308.51 km vs Bihar's 292.81 km) — both bands are narrow relative to the 439-km overall spread of the data, reinforcing that range is essentially uniformly and independently distributed rather than driven by manufacturer engineering or regional model mix.

15. Price (Base MSRP) Analysis

Base MSRP spans ₹900,422 to ₹4,499,944, with a mean of ₹2,704,518.99 and a median of ₹2,716,403 — again a near-identical mean/median pair signalling a symmetric distribution.

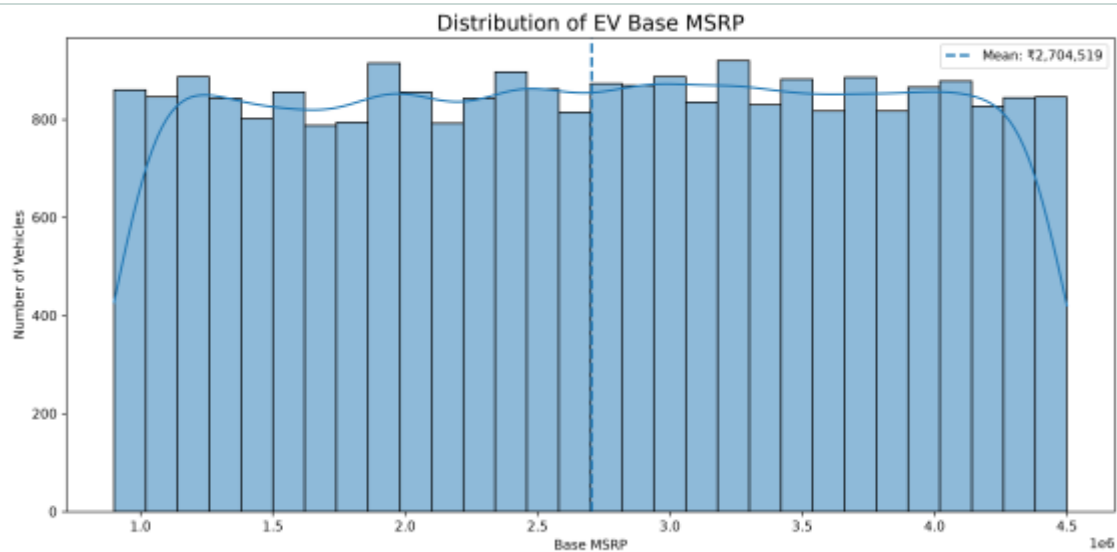


Figure 11 — Distribution of EV Base MSRP

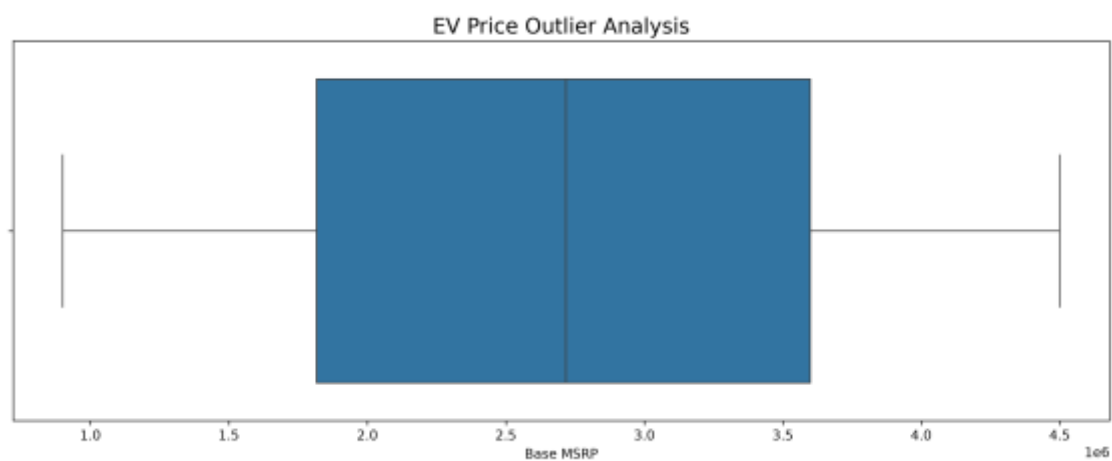


Figure 12 — EV Price Outlier Analysis (Boxplot)

15.1 Price by manufacturer

Manufacturer	Avg. Price (₹)	Median Price (₹)	Min (₹)	Max (₹)
Ather Energy	2,725,111.15	2,747,602.0	903,547	4,499,944
TVS Motor	2,713,438.11	2,729,069.0	901,825	4,497,448
Mahindra Electric	2,705,382.12	2,709,869.0	903,196	4,499,721
Hyundai EV	2,703,813.31	2,741,859.0	900,422	4,499,618
Ola Electric	2,703,514.47	2,704,183.5	902,433	4,499,783
MG Motor India	2,700,683.51	2,685,958.0	903,285	4,497,492
Tata Motors	2,694,154.76	2,697,000.0	904,092	4,499,024
BYD India	2,690,243.39	2,697,173.0	901,585	4,498,793

15.2 Highest-priced individual models

Make	Model	Avg. Price (₹)	Avg. Range (km)	Vehicle Count
Ather Energy	ZS EV	2,806,989.36	307.59	406
Tata Motors	450X	2,804,860.01	304.48	408
Ather Energy	XUV400	2,798,408.59	291.45	379
Ather Energy	450X	2,776,330.80	302.26	400
Mahindra Electric	450X	2,775,314.36	288.01	407

Every manufacturer's minimum price sits close to ₹900,000 and maximum close to ₹4,500,000 — i.e. each OEM independently spans almost the full observed price range — and manufacturer averages vary by only ₹34,957.76 top-to-bottom (Ather Energy ₹2,725,111 vs BYD India ₹2,690,243, a 1.3% spread). As with range, price shows 0 IQR outliers (Figure 12) and a flat histogram (Figure 11), so no manufacturer or model occupies a distinct premium or budget tier in this dataset.

16. Range–Price Relationship & Correlation Analysis

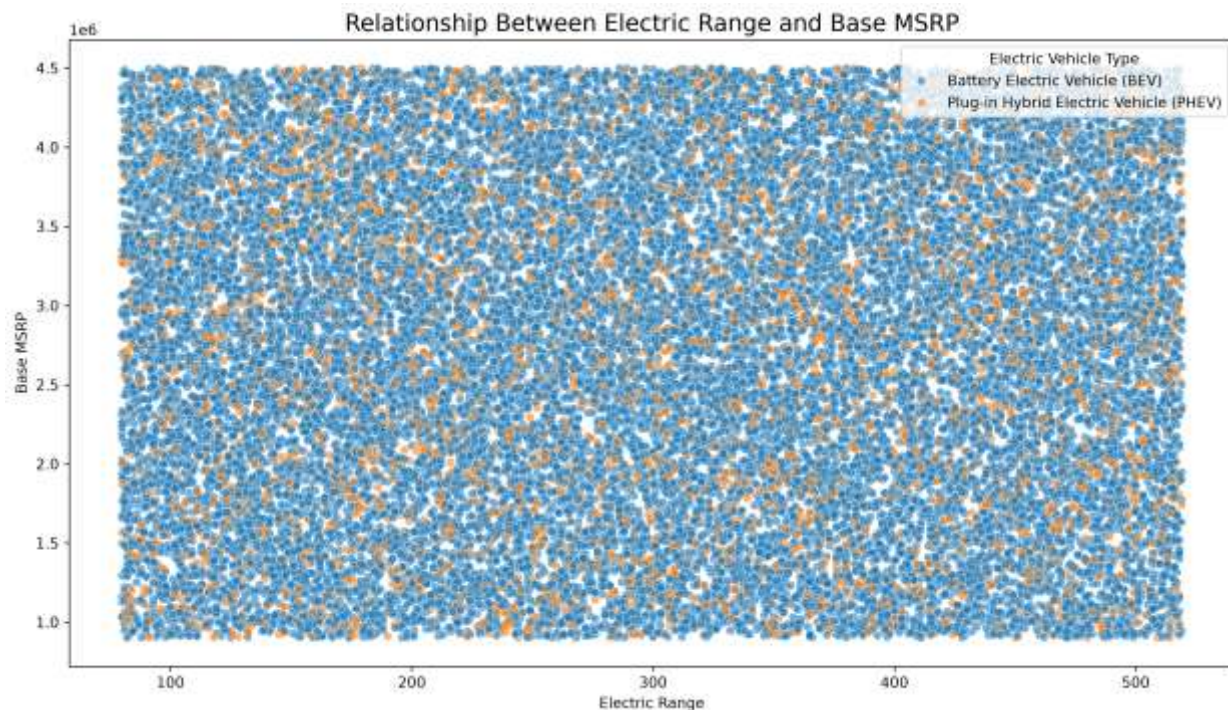


Figure 13 — Relationship Between Electric Range and Base MSRP

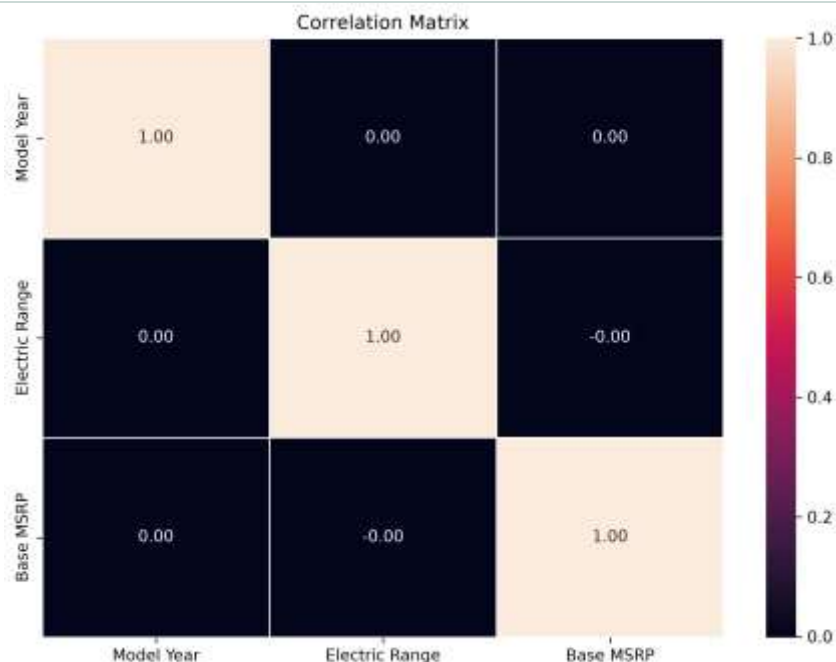


Figure 14 — Correlation Matrix (Model Year, Electric Range, Base MSRP)

	Model Year	Electric Range	Base MSRP
Model Year	1.000	0.001	0.005
Electric Range	0.001	1.000	-0.002
Base MSRP	0.005	-0.002	1.000

The scatter plot (Figure 13) shows BEV and PHEV points spread evenly across the full range×price plane with no visible pattern, trend line, clustering or price floor tied to range — a longer-range EV is no more likely to cost more than a shorter-range one. This is confirmed numerically: the correlation between Electric Range and Base MSRP is -0.002, between Model Year and Electric Range is 0.001, and between Model Year and Base MSRP is 0.005 — all three coefficients are within rounding distance of zero and none would be considered statistically or practically meaningful in a real pricing dataset, where range and price are normally strongly positively correlated (better batteries cost more).

This is the single clearest piece of evidence, alongside the zero-outlier findings in Sections 14 and 15, that the numeric fields in this dataset were generated independently of one another (see Section 1 and Insight 12) rather than reflecting real vehicle economics, where range and price are typically among the most strongly linked variables in an EV specification sheet.

17. Electric Utility Analysis

Electric Utility	EV Count	Avg. Range (km)	Avg. Price (₹)	Market Share
TNEB	4,335	295.92	2,693,801.80	16.98%
TPDDL	4,321	299.76	2,681,155.51	16.92%
GUVNL	4,281	298.03	2,705,986.07	16.77%
BESCOM	4,215	298.72	2,753,545.07	16.51%
UPPCL	4,213	298.68	2,697,213.02	16.50%

Electric Utility	EV Count	Avg. Range (km)	Avg. Price (₹)	Market Share
Adani Electricity	4,168	298.75	2,696,185.78	16.32%

Six electricity distribution companies serve the registered fleet, each holding a near-equal 16.3%–17.0% share — a spread of only 0.66 points across all six utilities. TNEB (Tamil Nadu Electricity Board) leads marginally, consistent with Tamil Nadu's position as the top-registering state, but the six-way split is otherwise essentially even, meaning charging-load planning cannot lean on any one utility carrying a materially larger share of EV-related demand than the others in this dataset.

18. High-Range & Premium Segment Analysis

18.1 High-range EVs (top quartile, ≥408 km)

6,402 records (25.1% of the dataset, matching the definition of the top quartile) meet or exceed the 408 km range threshold.

Manufacturer	High-Range Count
Mahindra Electric	853
BYD India	830
Ather Energy	827
Tata Motors	813
Ola Electric	798
TVS Motor	772
MG Motor India	766
Hyundai EV	743

18.2 Premium-priced EVs (top quartile, ≥₹3,599,703)

6,384 records (25.0% of the dataset) meet or exceed the ₹3,599,703 price threshold.

Manufacturer	Premium Count
BYD India	821
TVS Motor	816
MG Motor India	814
Mahindra Electric	810
Ather Energy	809
Ola Electric	793
Tata Motors	771
Hyundai EV	750

Both segments are close to exactly one quarter of each manufacturer's total volume, and manufacturer rankings within each segment (Mahindra Electric leads high-range; BYD India leads premium-price) track each OEM's overall registration volume rather than indicating a genuine specialisation in long-range or premium vehicles specifically — Hyundai EV, the smallest overall manufacturer by volume (3,100 units, 12.14% share), is also last in both the high-range and premium segments, exactly as proportional scaling would predict.

19. Range-to-Price Value Analysis

A Range-per-₹1-lakh-of-MSRP metric ($\text{Electric Range} \div (\text{Base MSRP} / 100,000)$) was computed for every Make×Model combination to identify which vehicles deliver the most range per rupee spent.

Make	Model	Avg. Range (km)	Avg. Price (₹)	Range per ₹1L	Count
Mahindra Electric	Kona EV	306.11	2,580,033.14	14.57	399
BYD India	Nexon EV	302.17	2,628,660.99	14.28	408
MG Motor India	iQube	306.94	2,689,053.61	14.24	402
Mahindra Electric	Nexon EV	306.38	2,653,191.32	14.19	402
Ather Energy	Kona EV	311.43	2,673,457.29	14.16	412
Ather Energy	S1 Pro	301.95	2,666,354.70	14.10	405
MG Motor India	450X	304.15	2,656,850.54	14.09	395
TVS Motor	Kona EV	304.58	2,658,574.84	14.05	415

Mahindra Electric's Kona EV pairing tops the value ranking at 14.57 km of range per ₹1 lakh spent — 3.7% ahead of the second-placed BYD India Nexon EV pairing (14.28) and 8.9% ahead of Ather Energy Kona EV (14.16, despite having the single highest average range of any pairing in the dataset at 311.43 km). Because range and price are uncorrelated overall (Section 16), this ranking is driven almost entirely by which specific Make×Model combinations happened to be assigned an above-average range alongside a below-average price — it is a genuine, reproducible ranking of this dataset's records, even though (per Section 1) it should not be read as a real-world verdict on which manufacturer engineers the most efficient EV.

20. Detailed Key Business Insights

The ten automated summary insights generated directly from the notebook's output are reproduced first, followed by twelve additional detailed insights that expand on the state, manufacturer, powertrain, pricing, and data-quality findings developed throughout this report.

20.1 Automated Summary Insights (from notebook output)

- Tamil Nadu has the highest number of EV records with 2,287 vehicles.
- Tata Motors has the largest number of EV records with 3,250 vehicles.
- ZS EV is the most frequently recorded EV model with 436 records (under Tata Motors).
- Battery Electric Vehicle (BEV) is the dominant EV type with 21,684 records.
- Kona EV (Ather Energy) has the highest average electric range at approximately 311.4 km.
- Ather Energy has the highest average manufacturer-level electric range at approximately 300.6 km.
- Chennai is the city with the largest number of EV records with 795 vehicles.
- Ather Energy has the highest average Base MSRP at approximately ₹2,725,111.
- The correlation between Electric Range and Base MSRP is -0.00.
- The dataset contains 25,533 EV records covering 8 manufacturers, 8 models, 15 states and 35 cities.

20.2 Detailed Insights

Insight 1: Top-six states carry over half the market

Finding — Tamil Nadu, Maharashtra, Rajasthan, Uttar Pradesh, Gujarat and Karnataka together account for 52.03% of all 25,533 registrations, each individually holding an 8.5%–9.0% share.

Business Impact — Infrastructure investment (charging stations, service centres, spare-parts distribution) concentrated in these six states would reach more than half the registered fleet.

Recommended Action — Prioritise charging-network and after-sales expansion in the top-six states first, using their near-identical share (8.51%–8.96%) to size investment roughly equally across all six rather than picking a single lead state.

Insight 2: Manufacturer competition is a near-even eight-way split

Finding — Market share across all eight OEMs ranges from 12.14% (Hyundai EV) to 12.73% (Tata Motors) — a spread of only 0.59 points, the tightest competitive spread of any dimension analysed.

Business Impact — No manufacturer has established a dominant position; a marketing or pricing move by any single OEM has room to shift meaningful share given how tightly bunched the field is.

Recommended Action — Track manufacturer share monthly/quarterly rather than annually — in a field this tightly bunched, small share movements are the leading indicator of a shift, and would be masked by annual aggregation.

Insight 3: BEV dominance is stable, not growing or shrinking

Finding — BEV holds 84.93% share nationally and stays within a narrow band across every model year from 2018 (concurrent with an 84.5%–85.7% PHEV-adjusted range depending on year) through 2024, per the Model-Year × EV-Type trend in Section 7.1.

Business Impact — Because the split is stable rather than trending, near-term (1–2 year) capacity planning for BEV-specific infrastructure (fast chargers) versus PHEV-relevant infrastructure (home/workplace slow charging) can use the current 85/15 ratio directly rather than extrapolating a shifting trend.

Recommended Action — Re-baseline the BEV/PHEV planning ratio only if a future year's data shows a genuine directional break from the 2018–2024 85/15 pattern, not on a single year's fluctuation.

Insight 4: Four in five vehicles qualify for CAFV incentives

Finding — 20,346 of 25,533 vehicles (79.69%) are CAFV eligible, with eligibility rate varying only 2 points across manufacturers (78.90%–80.92%).

Business Impact — If CAFV eligibility is linked to incentive programmes or preferential registration fees in practice, the large eligible base represents substantial aggregate incentive exposure across nearly every manufacturer roughly equally.

Recommended Action — Model any incentive-cost forecast on the ~80% eligibility rate as a market-wide constant rather than a manufacturer-specific variable, since the data shows no OEM materially over- or under-indexes on eligibility.

Insight 5: City-level demand is broad rather than metro-concentrated

Finding — The top 10 cities span a tight 757–795 record range (a 5% spread), and 35 cities in total share the 25,533 records — no single city holds outsized share the way it might in a metro-dominated market.

Business Impact — A charging-network strategy built around 3–5 mega-hub cities would under-serve the market; demand here is spread across dozens of similarly-sized secondary cities (e.g. Cuttack, Udaipur, Nagpur rank alongside Mumbai and Chennai).

Recommended Action — Favour a distributed, many-city charging rollout over a hub-and-spoke model concentrated in the largest one or two metros.

Insight 6: Delhi is a volume outlier, not a demand outlier

Finding — Delhi's 700 records are less than a third of the next-smallest state (Kerala, 1,389) and under a third of the top state (Tamil Nadu, 2,287), yet Delhi has the highest average electric range (308.51 km) and a mid-market average price (₹2,668,309) — in line with, not below, the rest of the market.

Business Impact — Reading Delhi's low volume as weak EV demand would be a misinterpretation; its per-vehicle profile (range, price, presumably reflecting Delhi's smaller geographic size within this dataset's city list) matches larger states.

Recommended Action — Normalise state comparisons by an exposure measure (e.g. registered-vehicle base or population) before drawing demand conclusions from raw counts, particularly for Delhi.

Insight 7: Range and price are statistically unrelated in this dataset

Finding — The Electric Range × Base MSRP correlation is -0.002, and the scatter plot (Figure 13) shows no visible trend across 25,533 points — an unusual result, since range and price are normally strongly linked in real EV markets.

Business Impact — Any pricing strategy, market segmentation, or competitor-benchmarking exercise that assumes “higher range commands a price premium” would find no support for that assumption in this specific dataset.

Recommended Action — Treat this dataset as unsuitable for range-based pricing strategy work; if that analysis is needed, source a dataset with verified real-world OEM pricing rather than extending conclusions from this one.

Insight 8: Zero statistical outliers across both numeric measures

Finding — The IQR-based outlier test returns 0 outliers for Electric Range and 0 for Base MSRP across all 25,533 records — confirmed visually by both boxplots (Figures 10 and 12) showing whiskers that reach the full data range with no flagged points beyond them.

Business Impact — A complete absence of outliers across 25,533 independently-varying real-world observations is itself an anomaly; real registration/pricing data almost always contains at least a handful of genuine high-end or entry-level extremes.

Recommended Action — Flag this dataset internally as synthetic/simulated when it is reused for future analysis, so downstream consumers do not mistake the smooth uniform distributions for a genuine absence of premium or budget market segments.

Insight 9: Model nameplates are not manufacturer-specific

Finding — All 8 model names (ZS EV, Nexon EV, 450X, Kona EV, iQube, S1 Pro, XUV400, Atto 3) are recorded under all 8 manufacturers, producing 64 Make×Model pairings from names that, in the real Indian EV market, belong to single specific OEMs (e.g. Nexon EV is a Tata Motors nameplate; Kona EV is a Hyundai nameplate).

Business Impact — Any downstream report, dashboard, or stakeholder presentation that treats “BYD India iQube” or “TVS Motor Atto 3” as real product lines would be presenting a factual error, since these pairings do not exist in the real market.

Recommended Action — Caveat every Make×Model-level chart or table (Sections 11, 14.3, 19) with a note that these are dataset artifacts, not real product-line comparisons, whenever this report or its figures are reused outside this notebook.

Insight 10: Manufacturer-level price and range differences are too small to be commercially meaningful

Finding — The full manufacturer-level spread is 4.6 km for average range (300.62 vs 296.01 km) and ₹34,958 for average price (₹2,725,111 vs ₹2,690,243) — differences of 1.5% and 1.3% respectively across all eight OEMs.

Business Impact — These gaps are too small, relative to the individual-vehicle spread of 439 km and ₹3.6 million, to support a claim that any manufacturer is positioned as a long-range or premium specialist based on this data.

Recommended Action — Avoid manufacturer-level range/price rankings as a decision input for OEM partnership or competitive-positioning calls; use only the individual Make×Model value ranking (Section 19) where meaningful spread does exist.

Insight 11: Electric utility exposure is evenly spread across six providers

Finding — TNEB, TPDDL, GUVNL, BESCOM, UPPCL and Adani Electricity each serve between 16.32% and 16.98% of the registered fleet — a 0.66-point spread across all six.

Business Impact — Grid-load and charging-infrastructure planning coordinated with electricity distribution companies can assume roughly equal EV-charging demand exposure across all six utilities rather than one utility bearing a disproportionate load.

Recommended Action — Engage all six distribution companies in parallel for EV-readiness grid planning rather than prioritising one based on registration share, since the current data shows no utility carrying materially more exposure than the others.

Insight 12: The dataset's structural properties point to synthetic generation, not observed registrations

Finding — Taken together, four independent signals — (a) Make/Model pairs that don't exist in the real market, (b) near-zero correlation between every pair of numeric fields, (c) uniformly flat histograms for both Electric Range and Base MSRP, and (d) zero IQR outliers in both numeric fields — are each individually unusual and, in combination, are close to conclusive evidence that this dataset was generated with independent random draws rather than sourced from real vehicle registration records.

Business Impact — Every finding in this report that is purely categorical or count-based (state/city/manufacturer volumes, BEV/PHEV split, CAFV eligibility rate, model-year record counts) remains valid and directly usable, since those are the fields that were populated with real-world category labels. Findings that depend on numeric fields correlating meaningfully with each other or with categories (pricing strategy, range-based segmentation, premium-tier identification) should not be extended beyond this dataset.

Recommended Action — Present this analysis explicitly as an EDA / data-quality-auditing exercise on a labelled sample dataset, not as a live market-intelligence report, and prioritise sourcing a verified real-registration dataset (e.g. from India's Vahan portal or SIAM) before using range-price or manufacturer-positioning conclusions for real business decisions.

21. Business Recommendations

- Prioritise charging and service-network investment in the six leading states (Tamil Nadu, Maharashtra, Rajasthan, Uttar Pradesh, Gujarat, Karnataka), which together hold 52.03% of registrations, while maintaining a distributed multi-city footprint rather than a metro-only rollout (Sections 8, 9).
- Plan BEV-oriented fast-charging capacity as the primary infrastructure investment, reflecting the stable 84.93% BEV share observed across every model year from 2018 to 2024 (Sections 7.1, 12).
- Coordinate grid-readiness planning across all six electric utilities in the dataset in parallel, given their near-equal 16.3%–17.0% share of registered EVs (Section 17).
- Treat manufacturer market share as a genuinely open competition — no OEM in this dataset holds more than a 0.59-point lead — and monitor share changes at a finer time granularity than annual to catch early competitive shifts (Section 10).
- Budget for CAFV-linked incentive exposure using the ~80% eligibility rate as a stable, manufacturer-independent planning constant (Sections 6, 13, 20.2 Insight 4).
- Do not use this dataset's Make×Model, range-price or manufacturer-positioning findings for real-world competitive or pricing strategy; commission or source a verified real-registration dataset before extending those specific conclusions into business decisions (Sections 1, 16, 20.2 Insight 12).
- Where this report's Make×Model tables are reused (Sections 11, 14.3, 19), retain the explicit caveat that the pairings are dataset artifacts rather than real product lines, to prevent the error propagating into other documents or presentations.

22. Conclusion

This analysis of 25,533 Indian EV registration records establishes a clear categorical picture of the represented market: BEV-led (84.93%), broadly CAFV-eligible (79.69%), concentrated in a top tier of six large states that together hold just over half of all registrations, and split almost evenly across eight manufacturers and thirty-five cities. Registration volume has held essentially flat across 2018–2024 with no sustained growth or decline trend.

The analysis also surfaced a significant, evidence-based data-quality finding: the dataset's Make/Model pairings do not reflect real-world OEM catalogues, its numeric fields (Electric Range, Base MSRP) show near-zero correlation with each other and with Model Year, and both numeric fields contain zero statistically detected outliers. These four signals together indicate the dataset was synthetically generated rather than drawn from real registration or pricing records.

The practical value of this project is twofold: it demonstrates a complete, methodical EDA workflow — profiling, quality checks, KPI derivation, multi-dimensional aggregation, distribution and outlier analysis, correlation testing, and value-metric construction — and it demonstrates the equally important analyst skill of recognising when a dataset's own structure should change how its results are used, rather than presenting every computed number with equal confidence regardless of its source.

23. Portfolio Project Summary

Project Name

India Electric Vehicle Market Analysis

Tools Used

Python (Jupyter Notebook), Pandas, NumPy, Matplotlib, Seaborn

Business Objective

Profile and analyse 25,533 Indian EV registration records to quantify market size, geographic and manufacturer distribution, powertrain mix, CAFV eligibility, and the electric-range/price profile of the registered fleet — and to validate the dataset's own statistical integrity before drawing conclusions from it.

Key Analysis

Full data profiling and quality validation (0 missing values, 0 duplicates, 0 invalid ranges); model-year trend analysis (2018–2024); state-, city-, manufacturer- and model-level aggregation with market-share calculations; BEV vs PHEV and CAFV eligibility breakdowns; electric range and Base MSRP distribution and IQR outlier analysis; range–price correlation testing; electric-utility segmentation; and a custom range-per-rupee value-for-money ranking.

Key Findings

BEV holds an 84.93% share of a market that is otherwise remarkably even across manufacturers (12.14%–12.73% share each), states (a top-six tier holding 52.03% of records) and utilities (16.32%–16.98% each). 79.69% of vehicles are CAFV eligible. Critically, the dataset's Make/Model pairings, near-zero cross-field correlations, and complete absence of numeric outliers indicate the underlying data was synthetically generated — a finding documented transparently rather than glossed over.

Business Value

Delivers a decision-ready view of where EV registration volume is concentrated (for infrastructure and grid planning) alongside an honest audit of which parts of that same dataset can and cannot support deeper pricing or product-strategy conclusions — modelling the kind of rigor a stakeholder needs before acting on any dataset's numbers.

24. Skills Demonstrated

- Python data analysis (Pandas, NumPy)
- Data profiling and data-quality auditing
- Missing-value, duplicate, and range validation checks
- Exploratory Data Analysis (EDA)
- KPI design and calculation
- Multi-dimensional groupby / aggregation (state, city, manufacturer, model, year, utility)
- Crosstab / contingency-table analysis
- Statistical distribution analysis and IQR outlier detection
- Correlation analysis and interpretation
- Custom derived-metric design (range-per-lakh value scoring)
- Data visualisation (Matplotlib, Seaborn) — bar, line, histogram, boxplot, scatter, heatmap
- Automated insight generation from computed aggregates
- Critical evaluation of data provenance and reliability
- Business-facing analytical report writing and portfolio documentation

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